

ADVANCE PROBLEM PROGRAMMING

SLO 1: Functional Programming Concepts

1. Create a Pure Function in Java Using Function<T,R>

A **pure function** always produces the same output for the same input and does not modify any external data.

```
import java.util.function.Function;

public class Main {

    public static void main(String[] args) {

        Function<Integer, Integer> square = n -> n * n;

        System.out.println("Square = " + square.apply(5));

    }

}
```

Output:

Square = 25

The square function is pure because it does not change any external data and always returns the same result for the same input.

2. What is a Higher-Order Function? Give a Java Example.

A **higher-order function** is a function that takes another function as an argument or returns a function as its result.

Java supports higher-order functions using **functional interfaces** and **lambda expressions**.

```
import java.util.function.Function;

public class Main {
```

```
static int calculate(int n, Function<Integer, Integer> operation) {  
    return operation.apply(n);  
}  
  
public static void main(String[] args) {  
    Function<Integer, Integer> square = x -> x * x;  
  
    int result = calculate(5, square);  
  
    System.out.println("Result = " + result);  
}  
}
```

Output:

Result = 25

Here, calculate() is a higher-order function because it accepts another function, square, as a parameter.

3. Show Immutability Using final Variables in Java

Immutability means that a value cannot be changed after it has been assigned.

In Java, the final keyword prevents a variable from being reassigned.

```
public class Main {  
    public static void main(String[] args) {  
        final int number = 100;
```

```
        System.out.println("Number = " + number);  
    }  
}
```

Output:

Number = 100

The following statement would cause a compilation error:

```
number = 200;
```

because number is declared as final.

Therefore, final variables help achieve immutability by ensuring that the variable cannot be assigned a new value after initialization.